26 January 1987
USED IN LIEU OF
MIL-C-85049/55
16 January 1980

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, 90 , SELF-LOCKING, CATEGORY 4C (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3 AND MIL-C-83723 SERIES III CONNECTORS)

This military specification has been prepared by the Electronic Support Division of Air Force Logistics Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-85049/55. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the accessories described herein shall consist of this specification and the latest issue of MIL-C-85049.

This document covers self-looking parts only. Non self-looking parts may be acquired under MIL-C-85049/55 and are acceptable for use in USAF systems.

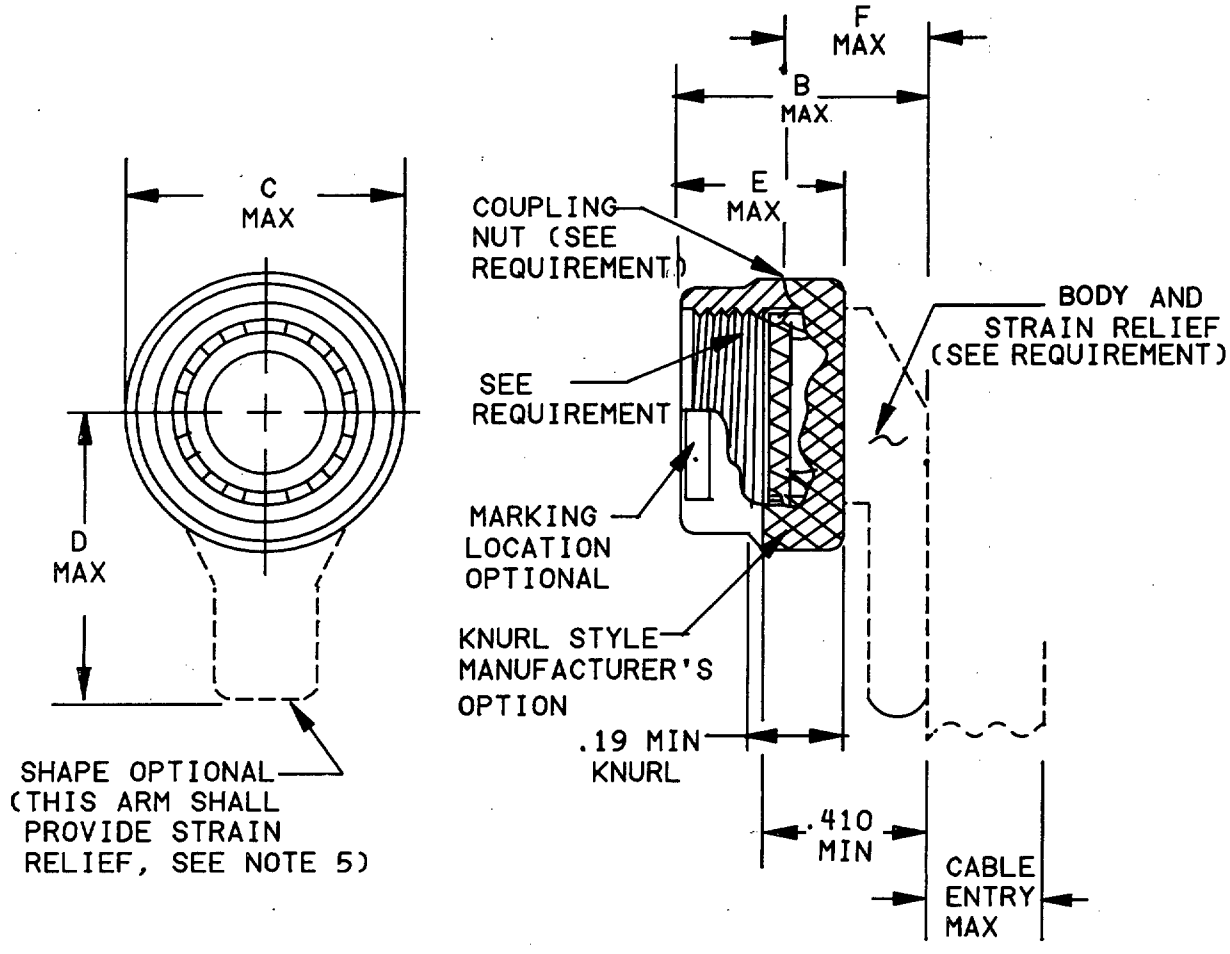


FIGURE 1. Dimensions and configuration.

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Dash numbers, shell size and dimensions

Dash number	For connector shell size (ref)				B max	C dia max	D max	E max	Cable entry max	F max
	MIL-C-81703 series 3	MIL-C-26482 series 2	MIL-C-5015 crimp	MIL-C-83723 series III						
03	3				.94 (23.9)	.885 (22.48)	1.29 (32.8)	.710 (18.03)	.260 (6.60)	.61 (15.5)
08		8	8S	8	.94 (23.9)	.885 (22.48)	1.22 (31.0)	.710 (18.03)	.260 (6.60)	.61 (15.5)
10		10	10S, 10SL	10	.94 (23.9)	1.010 (25.65)	1.29 (32.8)	.710 (18.03)	.365 (9.27)	.61 (15.5)
12	7	12	12, 12S	12	.94 (23.9)	1.135 (28.83)	1.62 (41.1)	.710 (18.03)	.501 (12.73)	.61 (15.5)
14	12	14	14, 14S	14	1.01 (25.7)	1.260 (32.00)	1.66 (42.2)	.710 (18.03)	.575 (14.61)	.68 (17.3)
16	19	16	16, 16S	16	1.01 (25.7)	1.385 (35.18)	1.72 (43.7)	.710 (18.03)	.700 (17.78)	.68 (17.3)
18	27	18	18	18	1.01 (25.7)	1.510 (38.35)	1.72 (43.7)	.710 (18.03)	.779 (19.79)	.68 (17.3)
20	37	20	20	20	1.07 (27.2)	1.635 (41.53)	1.79 (45.5)	.710 (18.03)	.904 (22.96)	.74 (18.8)
22		22	22	22	1.07 (27.2)	1.760 (44.70)	1.85 (47.0)	.710 (18.03)	1.029 (26.14)	.74 (18.8)
24		24	24	24	1.07 (27.2)	1.885 (47.88)	1.91 (48.55)	.710 (18.03)	1.144 (29.06)	.74 (18.8)
28			28		1.24 (31.5)	2.135 (54.23)	2.06 (52.3)	.890 (22.60)	1.379 (35.03)	.74 (18.8)

NOTES:
1. Dimensions are in inches.
2. Tolerances shall be: xx = ±.03 and .xxx = ±.015 angular tolerances x = ±2.
3. Metric equivalents are given for general information only.
4. Millimeters are in parentheses.
5. Strain relief unit is used with MS3367-1-X plastic strap, MIL-T-713 lacing twine or MIL-T-43435 lacing tape or equivalent, not supplied with the backshell.
6. This accessory will not accommodate connectors using size 8, 4, or 0 contacts.

FIGURE 1. Dimensions and configuration - Continued.

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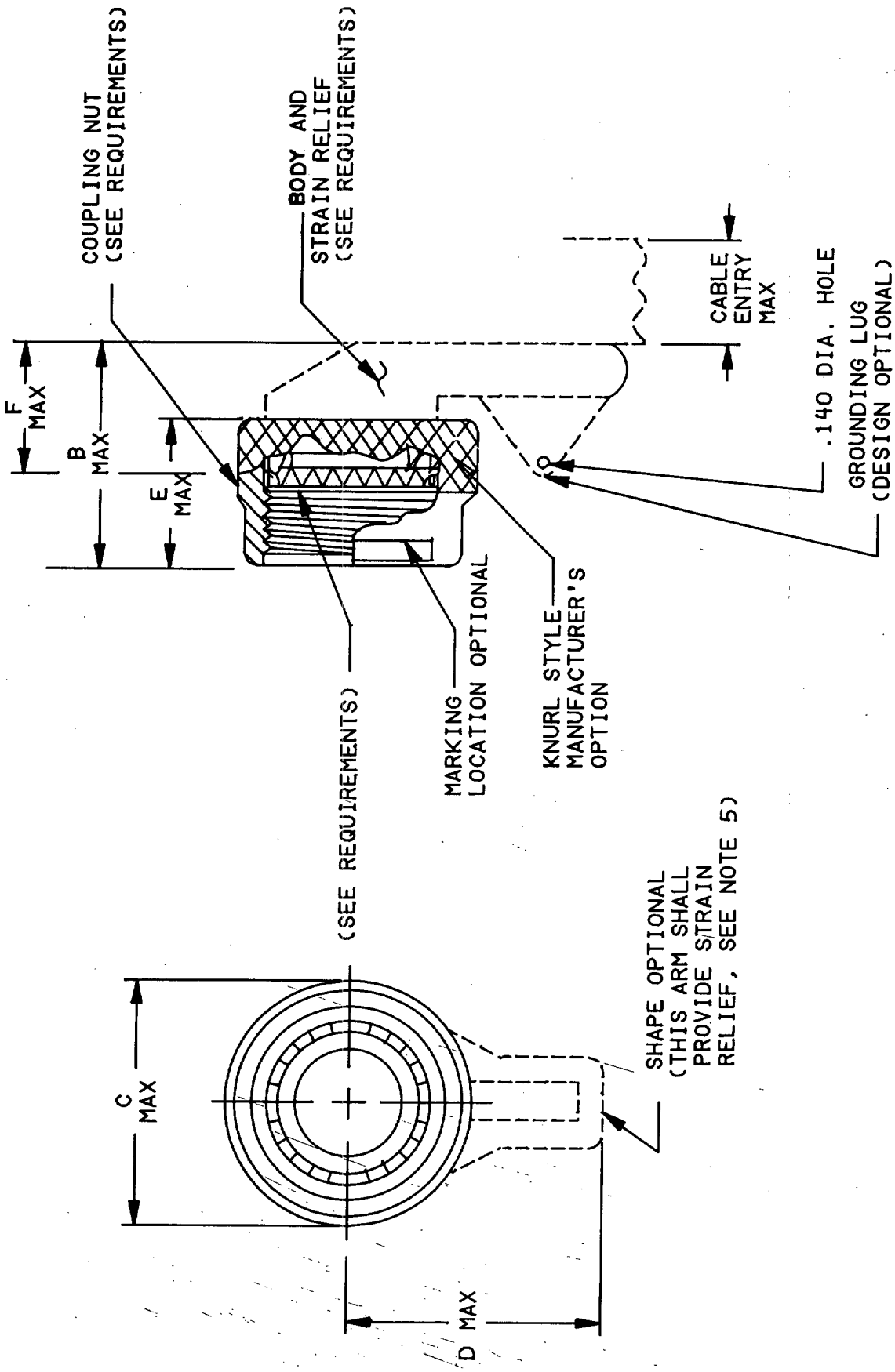


FIGURE 2. Dimensions and configuration with grounding lug.

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Dash numbers, shell size and dimensions

Dash number	For connector shell size (ref)				B max	C dia max	D max	E max	Cable entry max	F max
	MIL-C-81703 series 3	MIL-C-26482 series 2	MIL-C-5015 crimp	MIL-C-83723 series III						
03	3				1.05 (26.7)	.885 (22.48)	1.29 (32.8)	.710 (18.03)	.260 (6.60)	.61 (15.5)
08		8	8S	8	1.05 (26.7)	.885 (22.48)	1.22 (31.0)	.710 (18.03)	.260 (6.60)	.69 (17.5)
10		10	10S, 10SL	10	1.05 (26.7)	1.010 (25.65)	1.29 (32.8)	.710 (18.03)	.365 (9.27)	.69 (17.5)
12	7	12	12, 12S	12	1.05 (26.7)	1.135 (28.83)	1.62 (41.1)	.710 (18.03)	.501 (12.73)	.69 (17.5)
14	12	14	14, 14S	14	1.05 (26.7)	1.260 (32.00)	1.66 (42.2)	.710 (18.03)	.575 (14.61)	.69 (17.5)
16	19	16	16, 16S	16	1.05 (26.7)	1.385 (35.18)	1.72 (43.7)	.710 (18.03)	.700 (17.78)	.69 (17.5)
18	27	18	18	18	1.05 (26.7)	1.510 (38.35)	1.72 (43.7)	.710 (18.03)	.779 (19.79)	.69 (17.5)
20	37	20	20	20	1.07 (27.2)	1.635 (41.53)	1.79 (45.5)	.710 (18.03)	.904 (22.96)	.74 (18.8)
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24		24	24	24	1.07 (27.2)	1.885 (47.88)	1.91 (48.55)	.710 (18.03)	1.144 (29.06)	.74 (18.8)
28			28		1.24 (31.5)	2.135 (54.23)	2.06 (52.3)	.890 (22.60)	1.379 (35.03)	.74 (18.8)

NOTES:
1. Dimensions are in inches.
2. Tolerances shall be: xx = ±.03 and .xxx = ±.015 angular tolerances x = ±2.
3. Metric equivalents are given for general information only.
4. Millimeters are in parentheses.
5. Strain relief unit is used with MS3367-1-X plastic strap, MIL-T-713 lacing twine or MIL-T-43435 lacing tape or equivalent, not supplied with the backshell.
6. This accessory will not accommodate connectors using size 8, 4, or 0 contacts.
7. Not for Navy use. Not for new design for Air Force.

FIGURE 2. Dimensions and configuration with grounding lug - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2.

Accessories consist of a coupling nut and body with strain relief arm. The coupling nut shall be captivated to, and free to rotate on the body.

Interface dimensions shall conform to MS3155.

Material:

Coupling nut and follower: Aluminum alloy in accordance with MIL-C-85049.

Tie mount: Polyamide (optional).

Fastener: Stainless steel (optional).

Finish: In accordance with MIL-C-85049.

N 1/ W

1/ Not for Navy use. Air Force use is for space applications only.

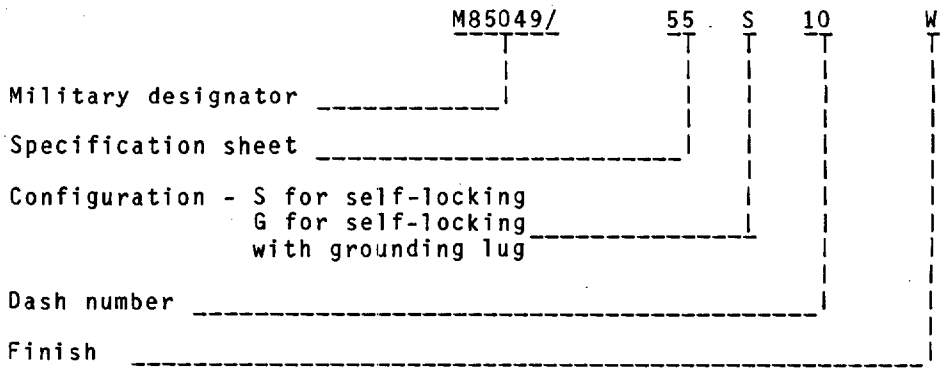
Self locking device within the coupling nut shall be corrosion resistant metal and shall provide a positive detent. Couplings with self-locking devices shall meet all the performance requirements specified herein for its specific category and the following additional tests.

Life cycle: The coupling with the locking device engaged shall be rotated sixty full turns in a clockwise direction and sixty full turns counter clockwise, prior to standard qualification testing. Cycle speed shall not exceed one full turn per second.

Locking: Coupling nut torque shall be 80 percent of coupling thread strength values given for medium and light duty accessories as specified in table III of MIL-C-85049. In table III when N/A appears, the values shall be 80 percent of those values used in qualification testing for coupling thread strength. Tolerance shall be ±5 inch pounds.

Vibration: Backshells shall be subjected to test parameters of applicable connector specifications. The coupling torque shall be within $^{+20}_{-10}$ inch pounds of the initial value.

Part number: See example below:



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QUALIFICATION:

See MIL-C-85049, category 4C, cable pull out not applicable.

QPL evaluating activity: Naval Avionics Center, Code B/714, 6000 East 21st Street, Indianapolis, Indiana 46219-2189.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Air Force - 85

Review activities:
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-F292-06)

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